

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<reasoning process here >

To find the de Broglie wavelength of an electron with a kinetic energy of1 eV , we use the de Broglie wavelength formula :

$\lambda = h / p$

where :

- h is Planck's constant = 6.626×10^{-34} J·s
- p is the momentum of the electron

For a non-relativistic electron , kinetic energy $K = (1/2)mv^2$, and momentum $p = mv$.

We can express momentum in terms of kinetic energy :

$p = \sqrt{2mK}$

So ,

$\lambda = h / \sqrt{2mK}$

We need to compute this with $K = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Mass of electron $m = 9.109 \times 10^{-31} \text{ kg}$

Plug in the values :

$\lambda = (6.626 \times 10^{-34} \text{ J}\cdot\text{s}) / \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$

First compute denominator :

$2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} = 2 \times 9.109 \times 1.602 \times 10^{-50} \mu\text{g}^\circ \approx 28.78 \times 10^{-50} \mu\text{g}^\circ = 2.878 \times 10^{-49} \text{ g}^\circ$

$\sqrt{2.878 \times 10^{-49} \text{ g}^\circ} \approx 5.365 \times 10^{-25} \mu\text{g}$

Now compute λ :

$\lambda = 6.626 \times 10^{-34} / 5.365 \times 10^{-25} \approx 1.235 \times 10^{-9} \text{ m}$

Convert to cm : $1.235 \times 10^{-9} \text{ m} = 1.235 \times 10^{-7} \text{ cm}$

So $\lambda \approx 1.235 \times 10^{-7} \text{ cm}$

Now look at the options :

- A. 2.48×10^{-8} cm $\approx$ too small
- B. 5.92×10^7 cm/sec $\approx$ not a wavelength
- C. 8.66×10^{-6} cm $\approx$ too large
- D. 3.14×10^{-5} cm $\approx$ too large
- E. 9.81×10^{-4} cm $\approx$ too large
- F. 5.27×10^{-9} m $\approx 5.27 \times 10^{-11}$ cm $\approx$ too small
- G. 1.23×10^{-7} cm $\approx$ matches our value
- H. 4.14×10^{-10} m $\approx 4.14 \times 10^{-8}$ cm $\approx$ too small
- I. 6.625×10^{-27} erg·sec $\approx$ Planck's constant in erg·sec , not wavelength
- J. 0.512×10^6 eV $\approx$ energy , not wavelength

So the correct answer is G : 1.23×10^{-7} cm

Note : 1.23×10^{-7} cm is equivalent to 1.23×10^{-9} m , which is a standard value for the de Broglie wavelength of a 1 eV electron .

</reasoning process>

<answer>G</answer>